

SCQ (Single Correct Type) :

- The value of  $\int x \cdot \frac{\ln(x + \sqrt{1+x^2})}{\sqrt{1+x^2}} dx$  equals:

(A)  $\sqrt{1+x^2} \ln(x + \sqrt{1+x^2}) - x + C$       (B)  $\frac{x}{2} \cdot \ln^2(x + \sqrt{1+x^2}) - \frac{x}{\sqrt{1+x^2}} + C$

(C)  $\frac{x}{2} \cdot \ln^2(x + \sqrt{1+x^2}) + \frac{x}{\sqrt{1+x^2}} + C$       (D)  $\sqrt{1+x^2} \ln(x + \sqrt{1+x^2}) + x + C$
- The value of  $2 \int \sin x \cdot \operatorname{cosec} 4x dx$  is equal to

(A)  $\frac{1}{2\sqrt{2}} \ln \left| \frac{1+\sqrt{2}\sin x}{1-\sqrt{2}\sin x} \right| - \frac{1}{4} \ln \left| \frac{1+\sin x}{1-\sin x} \right| + C$

(B)  $\frac{1}{2\sqrt{2}} \ln \left| \frac{1+\sqrt{2}\sin x}{1-\sqrt{2}\sin x} \right| + \frac{1}{4} \ln \left| \frac{1+\sin x}{1-\sin x} \right| + C$

(C)  $\frac{1}{2\sqrt{2}} \ln \left| \frac{1-\sqrt{2}\sin x}{1+\sqrt{2}\sin x} \right| - \frac{1}{4} \ln \left| \frac{1+\sin x}{1-\sin x} \right| + C$

(D) none of these
- The values of a and b which satisfy  $\int \frac{2\sin 2x - \cos x}{4 - \cos^2 x - 4\sin x} dx = a \log |\sin x - 1| + b \log |\sin x - 3| + c$ , are \_\_\_\_\_.

(A)  $-\frac{3}{2}, \frac{11}{2}$       (B)  $\frac{3}{2}, \frac{11}{2}$       (C)  $\frac{3}{2}, -\frac{11}{2}$       (D)  $-\frac{3}{2}, -\frac{11}{2}$
- Let  $\int \frac{g(x)((1+x)\ln x - 5)}{x(\ln x)^6} dx = \frac{g(x)}{(\ln x)^5} + c, g(1) = e$  and  $g: (0, e) \rightarrow (0, \infty)$ . If  $g^n(x)$  denotes the  $n^{\text{th}}$  derivative of  $g(x)$ , then the value of  $g^{2012}(1)$  is \_\_\_\_\_.

(A)  $2011g(1)$       (B)  $2012g(1)$       (C)  $2013g(1)$       (D)  $2014g(1)$

5.  $\int e^{\sin x} \left( \frac{x \cos^3 x - \sin x}{\cos^2 x} \right) dx =$
- (A)  $e^{\sin x} (x + \sec x) + c$  (B)  $e^{\sin x} (1 + \sec x) + c$   
 (C)  $e^{\sin x} (1 - \sec x) + c$  (D)  $e^{\sin x} (x - \sec x) + c$
6. If  $\int \frac{x dx}{\sqrt[2012]{(1+x^2)^{1012} (2+x^2)^{3012}}} = \frac{\alpha}{\beta} (1-f(x))^{\frac{\beta}{2\alpha}} + k$  then which is true
- (A)  $\alpha = 503; \beta = 500, f(\sqrt{2}) = \frac{1}{\beta - \alpha}$  (B)  $\alpha = 503; \beta = 250, f(\sqrt{2}) = \frac{1}{\alpha - \beta}$   
 (C)  $\alpha = 503; \beta = 500, f(1) = \frac{1}{\alpha - \beta}$  (D)  $\alpha = 503; \beta = 225, f(\sqrt{3}) = \frac{1}{\alpha - \beta}$
7. If  $\int \frac{\sec^2 x - 2010}{\sin^{2010} x} dx = \frac{P(x)}{\sin^{2010} x} + C$ , then value of  $P\left(\frac{\pi}{3}\right)$  is \_\_\_\_.
- (A) 0 (B)  $\frac{1}{\sqrt{3}}$  (C)  $\sqrt{3}$  (D) None of these
8. If  $g(x) = \frac{f(x)}{(x-a)(x-b)(x-c)}$  where  $f(x)$  is a polynomial of degree  $< 3$ , then \_\_\_\_.
- (A)  $\int g(x) dx = \begin{vmatrix} 1 & a & f(a) \log |x-a| \\ 1 & b & f(b) \log |x-b| \\ 1 & c & f(c) \log |x-c| \end{vmatrix} \div \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} + k$   
 (B)  $\frac{d}{dx} g(x) = \begin{vmatrix} 1 & a & f(a) \log(x-a)^{-2} \\ 1 & b & f(b) \log(x-b)^{-2} \\ 1 & c & f(c) \log(x-c)^{-2} \end{vmatrix} \div \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$   
 (C)  $\int g(x) dx = \begin{vmatrix} 1 & a & f(a) \log |x-a| \\ 1 & b & f(b) \log |x-b| \\ 1 & c & f(c) \log |x-c| \end{vmatrix} \div \begin{vmatrix} a^2 & a & 1 \\ b^2 & b & 1 \\ c^2 & c & 1 \end{vmatrix} + k$   
 (D) None of these
9. If  $y = 2 \cot^{-1} \left( \frac{x}{2} \right) + x$ , then  $\int \frac{x^2 dx}{(x^2 - 4) \sin x + 4x \cos x}$
- (A)  $\log |\operatorname{cosec} 2y + \cot y| + c$  (B)  $\log |\operatorname{cosec} 2y - \cot 2y| + c$   
 (C)  $\log |\operatorname{cosec} y + \cot y| + c$  (D)  $\log |\operatorname{cosec} y - \cot y| + c$

## Comprehension Type Question:

### Comprehension # 1

In calculating a number of integrals we had to use the method of integration by parts several times in succession. The result could be obtained more rapidly and in a more concise form by using the so-called generalized formula for integration by parts

$$\int u(x) v(x) dx = u(x) v_1(x) - u'(x) v_2(x) + u''(x) v_3(x) + \dots + (-1)^{n-1} u^{(n-1)}(x) v_n(x) - (-1)^{n-1} \int u^{(n)}(x) v_n(x) dx$$

$$\text{where } v_1(x) = \int v(x) dx, v_2(x) = \int v_1(x) dx, \dots, v_n(x) = \int v_{n-1}(x) dx$$

Of course, we assume that all derivatives and integrals appearing in this formula exist. The use of the generalized formula for integration by parts is especially useful when calculating  $\int P_n(x) Q(x) dx$ , where  $P_n(x)$  is polynomial of degree  $n$  and the factor  $Q(x)$  is such that it can be integrated successively  $n + 1$  times.

10. If  $\int (x^3 - 2x^2 + 3x - 1) \cos 2x dx = \frac{\sin 2x}{4} u(x) + \frac{\cos 2x}{8} v(x) + C$  then
- (A)  $u(x) = x^3 - 4x^2 + 3x$  (B)  $u(x) = 2x^3 - 4x^2 + 3x$   
(C)  $v(x) = 3x^2 - 4x + 3$  (D)  $v(x) = 6x^2 - 8x$
11. If  $\int (3x^2 + x - 2) \sin^2(3x + 1) dx = -\frac{u(x)}{72} \sin(6x + 2) + \frac{v(x)}{72} \cos(6x + 2) + \frac{1}{2} x^3 + \frac{1}{4} x^2 - x + C$
- (A)  $u(x) = 3x^2 + 6x - 13$  (B)  $u(x) = 18x^2 + 2x - 13$   
(C)  $v(x) = 3x + 1$  (D)  $v(x) = -(6x + 1)$

### Numerical based Questions :

12. If  $\int \frac{\sqrt{4+x^2}}{x^6} dx = \frac{(a+x^2)^{3/2} \cdot (x^2-b)}{120x^5} + C$  then  $a + b$  equals to :
13. Let  $g(x) = \int \frac{1+2\cos x}{(\cos x + 2)^2} dx$  and  $g(0) = 0$  then value of  $32g\left(\frac{\pi}{2}\right)$  is.
14. If  $f(x) = \sqrt{x-1}$  ;  $g(x) = e^x$  and  $\int f \circ g(x) dx = A f \circ g(x) + B \tan^{-1}(f \circ g(x)) + C$  then  $A^3 + B^2$  equals
15. If  $\int \frac{1+x\cos x}{x(1-x^2e^{2\sin x})} dx = k \ln \sqrt{\frac{x^2e^{2\sin x}}{1-x^2e^{2\sin x}}} + C$  then  $k$  is equal to :
16. Let  $\int \frac{f'(x)g(x) - g'(x)f(x)}{(f(x)+g(x))\sqrt{f(x)g(x)-g^2(x)}} dx = \sqrt{m} \tan^{-1}\left(\sqrt{\frac{f(x)-g(x)}{ng(x)}}\right) + C$  ,

where  $m, n \in \mathbb{N}$  and 'C' is constant of integration ( $g(x) > 0$ ). Find the value of  $(m^2 + n^2)$ .

### Matrix Match Type :

17. Let  $I = \int \frac{e^x}{e^{4x} + 1} dx$  and  $J = \int \frac{e^{-x}}{e^{-4x} + 1} dx$

Then for any arbitrary constant C, match the following

**Column-I**

**Column-II**

(A) I

(P)  $\frac{1}{\sqrt{2}} \tan^{-1} \left( \frac{e^{2x} - 1}{\sqrt{2}e^x} \right) + C$

(B) J + I

(Q)  $\frac{1}{2\sqrt{2}} \ln \left( \frac{e^{2x} - \sqrt{2}e^x + 1}{e^{2x} + \sqrt{2}e^x + 1} \right) + C$

(C) J - I

(R)  $\frac{1}{2\sqrt{2}} \left( \tan^{-1} \left( \frac{e^{2x} - 1}{\sqrt{2}e^x} \right) - \frac{1}{2} \ln \left( \frac{e^{2x} - \sqrt{2}e^x + 1}{e^{2x} + \sqrt{2}e^x + 1} \right) \right) + C$

(S)  $\frac{1}{2\sqrt{2}} \left( \tan^{-1} \left( \frac{e^{2x} - 1}{\sqrt{2}e^x} \right) + \frac{1}{2} \ln \left( \frac{e^{2x} - \sqrt{2}e^x + 1}{e^{2x} + \sqrt{2}e^x + 1} \right) \right) + C$

18.

**Column-I**

**Column-II**

(A)  $\int \frac{x^2(x^6 + x^5 - 1)dx}{(2x^6 + 3x^5 + 2)^2}$

(P)  $\frac{1}{6} \left( \frac{1}{x^{-3}} + \frac{3}{x^{-2}} \right)^{\frac{1}{2}} + C$

(B)  $\int \frac{(x^5 + x^4 + x^2)dx}{\sqrt{4x^7 + 5x^6 + 10x^4}}$

(Q)  $\frac{1}{2} (1 + x^{-2} + x^{-5})^{-2} + C$

(C)  $\int \frac{(2x^{12} + 5x^9)dx}{(x^5 + x^3 + 1)^3}$

(R)  $\frac{-1}{6} (2x^3 + 3x^2 + 2x^{-3})^{-1} + C$

(where C is the constant of integration.)

(S)  $x \left( \frac{x^3}{25} + \frac{x^2}{20} + \frac{1}{10} \right)^{\frac{1}{2}} + C$

### Subjective Type Questions :

19. Evaluate :  $\int \cos x \cdot e^x \cdot x^2 dx$

20. Evaluate :  $\tan^{-1} x \cdot \ln(1 + x^2) dx$ .

21. Evaluate :  $\int e^x \frac{1 + nx^{n-1} - x^{2n}}{(1 - x^n) \sqrt{1 - x^{2n}}} dx$

22. If  $\int \frac{x \cos \alpha + 1}{(x^2 + 2x \cos \alpha + 1)^{3/2}} dx = \frac{f(x)}{\sqrt{g(x)}} + C$  then find f(x) and g(x).

23.  $\int \frac{dx}{(\sin^2 x + 2 \cos^2 x)^2}$

24. Evaluate  $\int \frac{(x - \sin x)^{3/2}}{\sqrt{x}} \left\{ \frac{6x^2 \sin^2 \frac{x}{2}}{x - \sin x} + 3x \right\} dx$